

Reproducible Retrospective Evaluation of Clinical AI Governance Failures Using a Non-Compensatory Gate Framework

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Target Journal: NEJM AI

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Abstract

Background: Institutional AI governance frameworks, including NIST AI RMF, the EU AI Act, and ISO/IEC 23894, provide procedural guidance but do not operationalise quantitative, threshold-based evaluation with explicit pass/fail gate logic. Whether a non-compensatory (conjunctive) governance architecture—requiring independent minimum thresholds on every evaluation dimension—would have identified documented AI deployment failures has not been empirically tested against structured case evidence.

Methods

We developed a reproducible evaluation system comprising a five-gate non-compensatory governance engine, a structured evidence encoding pipeline with triangulated provenance tracking, and a benchmark dataset infrastructure with deterministic outputs and cryptographic hash-chain validation. The system processed structured evidence from 12 documented AI governance failures (spanning healthcare, criminal justice, education, employment, autonomous vehicles, and computer vision) drawn from 64 independent documentary sources. Feature vectors were encoded using a transparent rubric with confidence-weighted provenance classification. Ablation analysis, Monte Carlo perturbation testing, and compensatory scoring comparison were

conducted in parallel. An expanded provenance-tiered benchmark (91 cases across three encoding tiers) assessed scale generalisability.

Results

Under the moderate governance profile, the non-compensatory framework rejected 11 of 12 cases (sensitivity 91.7% in this retrospective sample). The safety gate was the most frequently binding constraint (10/12 cases, 83%). Every rejected case failed at least two independent gates (mean: 2.6 gate failures per rejection), demonstrating redundant multi-gate defence in depth. In two cases (Google Flu Trends, Uber AV), the non-compensatory framework rejected systems that compensatory scoring would have approved, because composite scores allowed strong evidence and traceability values to offset critical safety deficiencies. Governance outcomes were stable under Monte Carlo perturbation of evidence uncertainty bounds (12/12 cases, 100% stability across 200 iterations per case). The expanded benchmark rejected all 61 failure cases (100%) and approved all 30 FDA-authorised control devices, with the provenance tier structure making the evidential weight of each tier transparent.

Conclusions

A reproducible non-compensatory governance evaluation system flagged the substantial majority of documented AI deployment failures, with redundant gate coverage providing defence in depth. The two compensatory divergence cases illustrate how composite scoring can mask critical safety deficiencies. The system's deterministic pipeline, provenance-tracked encoding, and cryptographic reproducibility infrastructure provide the basis for independent replication and extension. A control cohort of 12 FDA-cleared AI-enabled medical devices, encoded under the identical 15-feature gate schema, yielded classification metrics against the 12 core failure cases of sensitivity 0.917 (11/12 failure cases blocked) and specificity 1.000 (12/12 controls approved). Under the expanded 91-case benchmark (61 failures, 30 controls), the framework achieved sensitivity 1.000 and specificity 1.000 with no misclassifications. Provenance-tier analysis revealed that this result is weighted toward lower-confidence encodings (Tier 1 mean confidence 0.591; Tier 2: 0.383; Tier 3: 0.380), and the canonical core-12 result (sensitivity 0.917) remains the authoritative primary metric. This result is derived from structured retrospective evaluation datasets and does not constitute prospective real-world validation.

Perfect separation under the expanded benchmark is an expected structural property of the non-compensatory architecture applied to cases with documented multi-gate governance deficiency, rather than a claim of generalised device safety. Prospective testing on independently encoded mixed cohorts remains necessary to stress-test these estimates.

Introduction

The deployment of artificial intelligence systems in high-stakes settings has produced a growing record of governance failures, from sepsis prediction models with poor real-world performance¹ to autonomous vehicle fatalities¹² and algorithmic grading controversies.⁹ These failures have motivated the development of structured governance frameworks intended to prevent unsafe or unethical AI deployments.

Decision theory raises a question directly relevant to AI governance: whether oversight should use compensatory scoring—where strength in one evaluation domain can offset weakness in another, as in multi-criteria decision analysis¹⁸ and composite indicator frameworks¹⁹—or non-compensatory (conjunctive) logic, where every domain must independently meet a minimum threshold. Compensatory approaches risk allowing systems with critical deficiencies to deploy if they score well on other criteria. Non-compensatory approaches enforce hard floors on every dimension but may be perceived as overly restrictive.

Several institutional frameworks (NIST AI RMF, EU AI Act, ISO/IEC 23894) address AI governance but do not implement threshold-based gate logic with deterministic outputs. This constitutes the operationalisation gap.

Simulation studies have demonstrated the theoretical properties of non-compensatory gates, showing that they achieve superior unsafe detection rates compared with compensatory scoring under a range of assumptions. However, these findings rest on synthetic systems drawn from assumed distributional models. In this study, we bridge the gap between simulation and structured case replay by presenting a reproducible evaluation system that processes structured evidence from 12 well-documented AI governance failures through a non-compensatory governance policy engine. We introduce ablation analysis to test the individual contribution of each gate and deploy a provenance-tiered expanded benchmark (91 cases) to assess scale generalisability.

This study provides the retrospective empirical test: whether the gate architecture would have flagged documented AI governance failures. The contribution of this work is threefold: a reproducible governance evaluation system with deterministic encoding, cryptographic hash-chain validation, and provenance tracking; a structured historical replay methodology that tests non-compensatory governance against documented deployment failures with triangulated evidence; and empirical evidence that non-compensatory gate logic provides redundant defence in depth against multi-dimensional governance deficiency.

Methods

This is a historical replay study applying a pre-specified governance policy engine to structured case evidence. The governance engine was developed as part of the Unified Dual-Mode Governance Engine (v2.0) and was not modified for this analysis. The replay mode shares the identical GovernancePolicyEngine and CompensatoryPolicyEngine as the companion Monte Carlo simulation mode, extending it with evidence ingestion, triangulated provenance tracking, evidence timing classification, feature dependency auditing, sensitivity analysis, and ablation analysis.

Case selection

We identified 12 cases of documented AI governance failures spanning 2014 to 2021. The primary analysis focuses on five healthcare cases (sepsis prediction, population health management, diabetic retinopathy screening, symptom checking, oncology treatment recommendation). Seven cross-sector cases (public health, criminal justice, employment, education, social media, computer vision, autonomous vehicles) are included as a secondary analysis to test whether multi-gate governance deficiency is domain-specific or domain-general. Cases were selected based on the availability of primary evidence from peer-reviewed publications, official regulatory reports, or authoritative investigative journalism. The sample is a convenience sample of well-documented failures and is not representative of all AI deployments.

Evidence encoding and triangulation

For each case, we extracted feature values for 15 structural causal model (SCM) variables from multiple independent documentary sources. Values were derived using a pre-specified numeric encoding rubric with documented anchors (for example, documented fatal incident mapped to

harm_severity of 1.0; peer-reviewed validation study with poor external performance mapped to intrinsic_safety of 0.35). Evidence was triangulated from 64 source objects across 12 cases, drawn from peer-reviewed studies, regulatory filings (including NTSB accident reports and Ofqual regulatory assessments), and investigative journalism (including ProPublica, Reuters, and STAT News).

The triangulation aggregator computed confidence-weighted means across independent sources for each feature, yielding 57 triangulated features and 123 passthrough features where only a single source was available. Source independence was enforced through atomic observation identifiers and phenomenon grouping to prevent double-counting when derivative reporting of the same empirical event was present. Each evidence source was classified by timing: whether it was available before deployment (pre-deployment, n=4) or discovered only after an incident (post-incident, n=60), enabling deployment-time versus forensic replay modes.

A feature dependency audit flagged 20 declared overlaps where features shared underlying evidence sources, consistent with the engine's known dependency matrix covering pairs such as (intrinsic_safety, stress_robustness) and (evidence_strength, evidence_visibility). Each encoded value was assigned a confidence level (0 to 1) and classified into one of four provenance classes: direct evidence (confidence 0.80 or above), rule-derived (0.50 to 0.79), imputed from context (0.25 to 0.49), or uncertain estimate (below 0.25). Plausible value ranges (low and high bounds) were computed for each feature based on confidence, enabling sensitivity analysis.

Governance policy engine

The governance engine implements five non-compensatory gates, each enforcing an independent minimum threshold. Gate 1 (Safety): observed intrinsic safety must meet the safety threshold. Gate 2 (Evidence): evidence strength must meet the evidence threshold. Gate 3 (Bias): bias harm index must not exceed the bias ceiling. Gate 4 (Calibration): uncertainty calibration must meet the calibration threshold. Gate 5 (Traceability): traceability integrity must meet the traceability threshold. Under non-compensatory logic, failure of any single gate blocks deployment approval. Four threshold profiles were evaluated: permissive, moderate, strict, and very strict. A compensatory policy engine computing a weighted composite score was applied in parallel for comparison.

Control cohort construction

A control cohort of 12 FDA-cleared AI devices was constructed for specificity estimation (Extended Data 5).

Core-equivalent evidence standardisation

Eight additional healthcare failure cases were upgraded to full methodological equivalence with the original 12 core cases using a structured evidence enrichment and triangulation process. For each case, triangulated feature values from 2-3 independent enrichment sources replaced imputed baseline values where the triangulated confidence exceeded the original. Pre-hardening values were preserved as an audit trail. Across 8 cases, 16 features were upgraded, 8 imputations removed, and 16 confidence scores improved. Two cases (COVID-19 CXR Shortcut Learning, Hidden Stratification) achieved encoding quality uplift from LOW to MODERATE.

Replay procedure and ablation

Each case was loaded, validated against the input schema, processed through the SCM structural equations, and evaluated against the five governance gates. The observation model noise layer from the simulation mode was bypassed because documented evidence constitutes the observation rather than a noisy signal of a latent trait. The procedure was repeated for low and high evidence bounds to assess sensitivity. Compensatory scoring was applied for comparison. For ablation analysis, we recomputed governance outcomes after removing each of the 5 single gates and all 10 pairwise gate combinations, assessing whether any case changed from REJECT to APPROVE.

Reproducibility infrastructure

The reproducibility infrastructure including schema validation, provenance tracking, and content-hashed identifiers is described in Extended Data 9.

The evidence enrichment pipeline is described in Extended Data 6.

Statistical analysis

Given the small sample ($N=12$) and the absence of a population sampling frame, we report descriptive statistics rather than inferential tests. Because the sample consists exclusively of

documented failures, we can estimate sensitivity (proportion of known failures flagged) but not specificity (the false-positive rate would require a cohort including confirmed safe deployments).

Results

Governance outcomes

Under the moderate governance profile, all 12 control devices were classified APPROVE (0/5 gates failed per device), producing the following confusion matrix against the 12 core failure cases: TP=11, FN=1, TN=12, FP=0 (sensitivity 0.917, specificity 1.000). This result is derived from structured retrospective evaluation datasets and does not constitute prospective real-world validation. The sole false negative (Google DR) passed all five gates with the narrowest margins in the cohort. Under the expanded 91-case benchmark, the framework rejected all 61 failure cases and approved all 30 controls (TP=61, FN=0, TN=30, FP=0; sensitivity 1.000, specificity 1.000). Three gates carried the primary discriminative load under both scopes: G1 (Safety), G4 (Calibration/Uncertainty), and G5 (Traceability/Monitoring).

Under the expanded benchmark, this perfect separation is an expected structural property of the non-compensatory architecture applied to cases with documented multi-gate governance deficiency. All 91 cases (61 failures, 30 controls) share a single canonical encoding schema with uniform feature definitions, triangulation classification, and gate traceability. The control cohort's feature values are encoded from FDA regulatory documentation with cross-cutting peer-reviewed validation sources. Under the core 12-case analysis, the single approved failure case (Google DR) illustrates the framework's behaviour at the decision boundary. Device-specific independent evaluations would further strengthen encoding quality differentiation across controls. These results should be interpreted as evidence that the framework discriminates between documented governance failures and regulatory-cleared systems, rather than as a robust specificity estimate.

Provenance-tier stratification of the expanded benchmark reveals an encoding quality gradient. Tier 1 (12 expert-triangulated core cases, mean confidence 0.591, 27.2% direct evidence) achieved sensitivity 0.917 under the canonical primary encoding and 1.000 under the expanded benchmark encoding. Tier 2 (49 additional documented failures, mean confidence 0.383, 1.9% direct evidence) achieved sensitivity 1.000. Tier 3 (30 FDA-cleared controls, mean confidence

0.380, 0% direct evidence) achieved specificity 1.000. The perfect separation under the expanded benchmark reflects the non-compensatory architecture's structural response to cases with documented multi-gate governance deficiency, amplified by the encoding quality gradient between failure cases and conservatively encoded controls. The canonical 11/12 result (sensitivity 0.917) remains the authoritative primary metric; the expanded result provides scale evidence at systematically lower evidential confidence.

The single approved case—Google's diabetic retinopathy screening system deployed in Thai community clinics—passed all five gates with the narrowest margins in the cohort. The safety gate margin was only 0.05 (observed 0.55 vs threshold 0.50). Google DR was the only case where the governance outcome was sensitive to feature perturbation: under sensitivity analysis (± 0.20 perturbation), 8 flip points were identified across 4 features (`intrinsic_safety`, `uncertainty_calibration`, `evidence_strength`, `traceability_integrity`), consistent with its narrow-margin approval status.

The safety gate was the most frequently binding constraint, with 10 of 12 cases (83%) breaching the safety threshold under the moderate profile (Figure 1). The bias gate failed in 6 cases (50%), reflecting documented subgroup disparity in Optum,² COMPAS,^{7,16} Amazon Recruiting,⁸ UK A-levels,⁹ Microsoft Tay,¹⁰ and Gender Shades.^{11,17} The calibration and traceability gates each failed in 5 cases (42%). The evidence gate failed in only 3 cases (25%), reflecting the deliberate selection of well-documented failures with substantial primary evidence. The mean number of gate failures per rejected case was 2.6 (29 total gate failures across 11 rejections), indicating that typical governance deficiency spans multiple evaluation dimensions.

Ablation analysis

Systematic ablation revealed that no single gate removal changed the governance outcome for any of the 12 cases (Figure 2A). Every rejected case failed at least two independent gates, demonstrating redundant multi-dimensional governance deficiency. Pairwise gate ablation identified three critical combinations (Figure 2B). Removing safety and bias simultaneously would have changed three rejections to approvals: Optum Population Health, Gender Shades, and UK A-level Algorithm—all characterised by documented subgroup disparities combined with marginal safety performance.

Removing safety and calibration would have approved Google Flu Trends and Uber AV, both combining safety deficiencies with poor uncertainty calibration. Removing evidence and traceability would have approved only Babylon Symptom Checker, the sole case whose rejection rests entirely on evidence and traceability weaknesses.

Non-compensatory versus compensatory comparison

The non-compensatory and compensatory engines agreed on 10 of 12 cases. In two cases, the non-compensatory framework rejected systems that compensatory scoring would have approved (Figure 4). Google Flu Trends was rejected by the safety and calibration gates despite scoring well on evidence strength (0.70) and traceability (0.40); its composite score of 0.57 exceeded the compensatory threshold of 0.50. The Uber AV case was rejected by the safety (0.10) and calibration (0.25) gates despite strong evidence (0.75) and traceability (0.70); its composite score of 0.51 crossed the compensatory approval line because high evidence and traceability scores offset the critically low safety value. No cases were approved by non-compensatory gates but rejected by compensatory scoring.

Evidence provenance and stability

Across all 12 cases and 15 features each (180 total feature encodings), 49 (27.2%) were classified as directly evidenced, 68 (37.8%) as rule-derived, 51 (28.3%) as imputed from context, and 12 (6.7%) as uncertain estimates (Figure 3A). Mean confidence across all encodings was 0.591 (median 0.600); 71 of 180 encodings (39.4%) had confidence ≥ 0.70 . The highest-confidence cases were those supported by multiple peer-reviewed or official regulatory sources (Uber AV, Epic Sepsis, Gender Shades). The lowest-confidence features were adversarial gaming capability (mean confidence 0.35 across all cases), reflecting the inherent difficulty of assessing gaming potential from documentary evidence.

Governance outcomes were robust to encoding uncertainty: Monte Carlo simulation sampling each feature value uniformly from its documented confidence bounds (200 iterations per case, deterministic seed = 42) produced stable outcomes for all 12 cases under the moderate profile (100% outcome stability; Figure 3B). Under sensitivity perturbation (± 0.20 on each gate feature independently), only the sole approved case (Google DR) showed outcome sensitivity, with 8 flip points across 4 features. All 11 rejected cases were insensitive to perturbation—no feature

shift within ± 0.20 changed any rejection, confirming that documented governance failures involved deficiencies well beyond threshold margins.

PhysioNet benchmark validation is reported in Extended Data 1.

PhysioNet benchmark models exhibited governance feature heterogeneity with different vulnerability patterns from historical failures (Extended Data 1).

Expanded benchmark evaluation

To assess whether the framework's sensitivity generalises beyond the 12 expert-encoded cases while maintaining explicit provenance accountability, we processed an expanded benchmark of 91 cases through the identical governance engine, organised into three provenance tiers based on encoding methodology.

Tier 1 comprised the 12 expert-triangulated cases described above (180 features from 64 documentary sources; mean confidence 0.591; 27.2% direct evidence, 37.8% rule-derived, 28.3% imputed from context, 6.7% uncertain estimate). Under the moderate profile, 11 of 12 were rejected (sensitivity 91.7%; Google DR approved with narrowest margins).

Tier 2 comprised 49 additional documented failure cases spanning peer-reviewed external validation studies, post-market surveillance analyses, subgroup harm studies, investigative reports, and fairness audits (2015–2026). Features were encoded from evidence metadata (failure mode class, evidence type, publication characteristics) using the same encoding anchors but with systematically lower provenance quality: 735 features at mean confidence 0.383, with 1.9% direct evidence, 11.7% rule-derived, 73.3% imputed from context, and 13.1% uncertain estimate. Under the moderate profile, all 49 were rejected (sensitivity 49/49), with a mean of 4.0 gate failures per rejection. The safety gate was binding for 100% of Tier 2 cases.

Tier 3 comprised 30 FDA-authorised AI-enabled medical devices encoded as specificity controls with conservatively favourable features derived from authorisation records (device name, manufacturer, submission number, intended use, authorisation date): 450 features at mean confidence 0.380, with 0% direct evidence, 13.3% rule-derived, 73.3% imputed, and 13.3% uncertain estimate. Under the moderate profile, all 30 were approved (specificity 30/30, preliminary).

The expanded benchmark (91 cases) is detailed in Extended Data 2.

Evidence enrichment and triangulation

Enriched triangulation evidence details are provided in Supplementary Appendix A.

Dual dataset structural invariance

To confirm that governance outcomes are determined by feature values rather than schema structure, a structurally normalised variant of the canonical dataset was constructed.

Normalisation comprised three operations: key reordering to conform to the public dataset schema, addition of schema-required fields absent in the canonical source, and alphabetical sorting of feature sub-keys. No feature values were altered; cryptographic hash verification confirmed value-level identity (SHA-256 of all 180 feature values: identical across both datasets). The deterministic evaluation engine was executed against both datasets under all four threshold profiles in replay mode, yielding 480 field-level comparisons ($12 \text{ cases} \times 4 \text{ profiles} \times 10 \text{ output fields}$). All 480 comparisons were identical: zero gate divergences, zero compensatory score differences, zero verdict changes. Schema-level transformations do not affect governance outcomes under deterministic evaluation.

A secondary analysis compared replay mode (5-gate evaluation only) with canonical full mode (5-gate evaluation plus SCM-derived abstention override and fallback safety check). Under the moderate profile, mode choice had zero effect on outcomes: all 12 cases produced identical verdicts across both modes. Under the permissive profile, three cases (Epic Sepsis, Google Flu Trends, Babylon) flipped from APPROVE to REJECT in canonical full mode because their SCM-derived abstention rates exceeded 0.5 while fallback safety was inadequate. This is consistent with the finding that the primary retrospective validation results (replay mode, moderate profile) are fully mode-invariant for this case set.

Perturbation robustness analysis

A controlled perturbation analysis tested governance outcome stability under adversarial feature shifts. Three features (`uncertainty_calibration`, `bias_harm_index`, `traceability_integrity`) were perturbed by ± 0.05 in the direction most likely to cause gate failure: calibration and traceability were decreased by 0.05, and bias was increased by 0.05. Safety and evidence features were held invariant. This represents the maximum single-step adversarial perturbation within the ± 0.05 envelope.

Under replay mode, 46 of 48 configurations (95.8%) were verdict-stable under the evaluated perturbation conditions. Two verdict flips occurred, both under the permissive profile only (Epic Sepsis and Babylon flipped APPROVE→REJECT due to new calibration and traceability gate failures). The moderate profile was fully stable: zero verdict flips across all 12 cases. Ten individual gate evaluations flipped across all profiles and cases, concentrated at three features sitting exactly on threshold boundaries: Google Flu traceability (0.40→0.35, crossing the 0.40 threshold), UK A-levels calibration (0.40→0.35), and Microsoft Tay traceability (0.40→0.35). Five compensatory decision flips occurred, including Uber AV under the moderate profile (score 0.5125→0.4875, crossing the 0.50 threshold). Google DR remained APPROVE under moderate perturbation, confirming its status as the narrowest-margin but still robust approval.

The perturbation results indicate that the non-compensatory gate architecture provides robust governance outcomes under small feature uncertainty.

Under conservative-bound analysis, in which all gate features were simultaneously set to their least favourable documented values (lower confidence bound for safety, evidence, calibration, and traceability; upper confidence bound for bias), all 12 governance verdicts were maintained. The sole approved case (Google DR, canonical encoding `intrinsic_safety` 0.55) survived at exact threshold boundaries (`intrinsic_safety` 0.50 against threshold 0.50; `uncertainty_calibration` 0.40 against threshold 0.40) under the engine's inclusive ($\geq$) gate logic. The 11 rejected cases remained well below their binding thresholds under conservative bounds, confirming that their rejection does not depend on marginal encoding judgements. The 100% moderate-profile verdict stability under adversarial perturbation strengthens the reliability of the primary 11/12 sensitivity finding. The compensatory decision flips (notably Uber AV crossing the 0.50 threshold) illustrate why compensatory scoring is more sensitive to small encoding perturbations than non-compensatory gating, consistent with the simulation findings in the companion study.

The interaction between parameter and evidence sensitivity dimensions is analysed in Extended Data 8.

Discussion

This study presents a reproducible evaluation system for AI governance and applies it in a structured historical replay against documented deployment failures. The central finding is that

under the moderate governance profile the non-compensatory framework rejected 11 of 12 well-documented failures under moderate governance stringency, achieving 91.7% sensitivity for known failures in this retrospective dataset, with the safety gate as the most frequently binding constraint.

Defence in depth

The ablation analysis yields a finding that strengthens the case for multi-gate governance: no single gate removal changes any rejection outcome. Every rejected case fails at least two independent gates (mean: 2.6 gate failures per rejection), indicating that the documented governance failures involved deficiencies across multiple evaluation dimensions. This redundancy—defence in depth—suggests that even if one gate were miscalibrated or bypassed, the remaining gates would have flagged each case. The pairwise ablation identifies the specific gate pairs whose joint failure defines each case's governance vulnerability: safety combined with bias (Optum, Gender Shades, UK A-levels), safety combined with calibration (Google Flu, Uber AV), and evidence combined with traceability (Babylon). These patterns are substantively coherent with the documented failure modes of each case.

The compensation effect

The compensatory policy engine applied in parallel with the non-compensatory framework provides an analytical comparator representing weighted additive aggregation—a standard approach in established multi-criteria decision analysis¹⁸ and health technology assessment methodologies.²¹

The two cases where non-compensatory and compensatory approaches diverged are particularly instructive. Both Google Flu Trends and Uber AV had strong evidence bases and reasonable traceability, but critical safety or calibration deficiencies. Compensatory scoring allowed these strengths to offset the safety weakness, producing an approval that the non-compensatory framework blocked. In both cases, the historical record is consistent with the interpretation that the safety concerns were material: Google Flu Trends suffered significant accuracy degradation,⁵ and the Uber AV system failed to identify a pedestrian, resulting in a fatality.¹²

The approved case

The single case approved under the moderate profile warrants examination. Google DR passed all five gates but with the narrowest margins observed. The approval should not be interpreted as an endorsement of the deployment: Beede et al.³ documented substantial operational challenges including poor image quality causing high rejection rates, workflow disruption, and connectivity failures. These implementation-level failures fall primarily within the operational domain rather than the intrinsic algorithmic properties captured by the SCM feature space. The governance evaluation system assesses pre-deployment algorithmic governance characteristics; operational deployment readiness assessment is a complementary but distinct function.

Healthcare implications

Healthcare-specific implications are discussed in Extended Data 10.

Relationship to existing frameworks

The non-compensatory gate approach tested here could function within NIST, EU AI Act, or ISO/IEC 23894 frameworks as the decision-theoretic engine underlying a conformity assessment or risk evaluation. The historical replay results suggest that explicit conjunctive gate structures may add value over procedural frameworks alone by enforcing quantitative floors that cannot be offset by strength in other domains—a non-compensatory constraint structure formalised in safety-critical systems engineering.²⁰

Scope and contribution

This work represents a structured historical replay evaluation and does not constitute prospective real-world validation. The contribution is threefold: the reproducible evaluation system itself, with its deterministic encoding pipeline, cryptographic hash-chain validation, and provenance-tracked benchmark infrastructure; the historical replay methodology that tests non-compensatory governance against documented deployment failures with triangulated evidence; and the empirical finding that non-compensatory gate logic provides redundant defence in depth against multi-dimensional governance deficiency.

Validation Hierarchy The evidence supporting the non-compensatory gate architecture currently spans two validation tiers: Tier 4 (computational simulation, characterising theoretical properties under modelled conditions) and Tier 3 (structured retrospective replay, testing the architecture against documented historical failures with triangulated evidence). Tier 2 (real-world

observational evaluation using institutional AI portfolio data) and Tier 1 (prospective controlled evaluation within live governance processes) remain outstanding and constitute the necessary next steps before deployment recommendations can be made. No finding in the present study should be interpreted as evidence from Tier 1 or Tier 2.

Patient-Level Consequence Pathway The governance failures replayed here produced documented patient-level consequences, including fatalities (Uber AV), inequitable healthcare allocation (Optum), and delayed diagnoses (Epic Sepsis). The non-compensatory framework evaluated in this study is designed to prevent the specific governance failure mode—compensatory masking of critical deficiencies—that contributed to these outcomes.

Scope Boundaries The gate architecture addresses governance evidence assessment within five specified domains. It does not address deployment risks arising from sociotechnical integration failures, procurement governance, clinical trial substitution where required by regulation, workflow integration, clinician trust calibration, or real-time human factors. These domains fall outside the architecture's scope and require complementary assessment processes.

Limitations

This study has several important limitations.

All evidence encoding was performed by a single author. An adversarial encoding perturbation analysis demonstrated that governance verdicts were stable under single-feature perturbation within a ± 0.10 stress-test envelope (239/240 non-baseline evaluations, 99.6%), and a conservative-bound analysis confirmed that all 12 verdicts were maintained under worst-case encoding values. These computational robustness checks demonstrate outcome insensitivity to single-feature perturbation within a ± 0.10 stress-test envelope but do not substitute for independent inter-rater reliability, which remains the highest-priority methodological next step. The ± 0.10 perturbation envelope is a stress-test bound, not an empirically calibrated measure of inter-coder disagreement.

First, the primary analysis uses 12 expert-triangulated cases as a curated convenience sample of well-documented failures. The expanded 91-case benchmark extends coverage across three explicit provenance tiers, but Tier 2 (49 cases) uses features derived within a unified encoding schema with variable provenance at substantially lower mean confidence (0.383 vs 0.591) and

near-absence of direct evidence (1.9% vs 27.2%), and Tier 3 (30 controls) uses imputed features with no direct evidence. The sample is not representative of all AI deployments, creating unavoidable selection and survivorship bias. Cases with thorough post-hoc investigation are overrepresented relative to cases that caused harm but were never scrutinised.

Evidence encoding methodology and limitations are detailed in Extended Data 17.

Additional limitations (retrospective design, sample size, control cohort) are detailed in Extended Data 17.

Ethics statement

This retrospective computational evaluation of publicly documented AI deployment cases does not involve human participants, patient data, or identifiable information. No ethics approval was required.

Funding

No external funding was received for this work.

Reporting guideline

This study was conducted with reference to the DECIDE-AI reporting framework for early-stage clinical AI evaluation.

Data availability

The historical replay artefact (12 expert-triangulated cases with 64 documentary evidence sources), expanded benchmark dataset (91 cases across three provenance tiers with 1,365 individually tracked feature encodings), governance engine source code, benchmark database, and reproducibility manifests are archived at Zenodo (DOI:

<https://doi.org/10.5281/zenodo.19388835>). The governance engine source code is available at <https://github.com/ethical-alpha-audit/ethical-alpha-audit-paper-4-historical-replay> under the MIT licence.

Code availability

The Unified Dual-Mode Governance Engine v2.0 and the AI Governance Benchmark Dataset System v1.0 are available at <https://github.com/ethical-alpha-audit/ethical-alpha-audit-paper-4->

historical-replay. The replay is fully deterministic and can be reproduced from clean setup. The benchmark dataset system includes schema validation, provenance tracking, stress testing, and content-hashed reproducibility manifests.

AI use disclosure

Claude (Anthropic) was used as a computational methods assistant for code generation, document production, and iterative review during the development of the historical replay system, benchmark dataset infrastructure, and manuscript preparation. All scientific decisions, evidence encoding judgements, and interpretive claims were made by the author. The governance policy engine and structural causal model were designed by the author prior to AI assistance. The AI assistant did not contribute to evidence selection, provenance classification, or encoding rule design.

Author contributions

W.B. conceived the study, designed the governance policy engine and structural causal model, developed the benchmark dataset system and reproducibility infrastructure, performed all evidence encoding and triangulation, conducted all analyses, and wrote the manuscript.

Competing interests

W.B. is the founder of Ethical Alpha Audit Ltd, which develops AI governance methodology. W.B. is also employed as an analyst at NHS England; all views expressed are the author's own and do not represent NHS England.

Figure legends

[Figure 1 near here]

Figure 1. Gate failure distribution and governance outcome matrix (moderate profile). Panel A: Number of cases (out of 12) failing each governance gate. Safety: 10 (83%); Bias: 6 (50%); Calibration: 5 (42%); Traceability: 5 (42%); Evidence: 3 (25%). Panel B: Governance outcomes across four threshold profiles (permissive, moderate, strict, very strict) for all 12 cases. Source: Governance Benchmark Dataset System v1.0 replay outputs.

[Figure 2 near here]

Figure 2. Ablation analysis (moderate profile). Panel A: Single gate removal—no single gate removal changes any outcome. Panel B: Pairwise gate removal—removing safety + bias changes 3 outcomes; safety + calibration changes 2; evidence + traceability changes 1.

[Figure 3 near here]

Figure 3. Evidence provenance, confidence distribution, and stability analysis. Panel A: Provenance class distribution across 180 feature encodings (27.2% direct evidence, 37.8% rule-derived, 28.3% imputed from context, 6.7% uncertain estimate). Mean confidence: 0.591. Panel B: Monte Carlo outcome stability: 12/12 cases stable under 200 iterations of uniform sampling within confidence bands. Sensitivity perturbation (± 0.20): only Google DR exhibits outcome sensitivity (8 flip points across 4 features).

[Figure 4 near here]

Figure 4. Compensation effect. Non-compensatory vs compensatory outcomes for all 12 cases. Two divergence cases (Google Flu composite score 0.57, Uber AV composite score 0.51) highlighted where compensatory scoring approved systems rejected by non-compensatory gates.

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Full reproducibility details are provided in Extended Data 9.

Extended scope conditions and supplementary methodological context are provided in Extended Data 7.